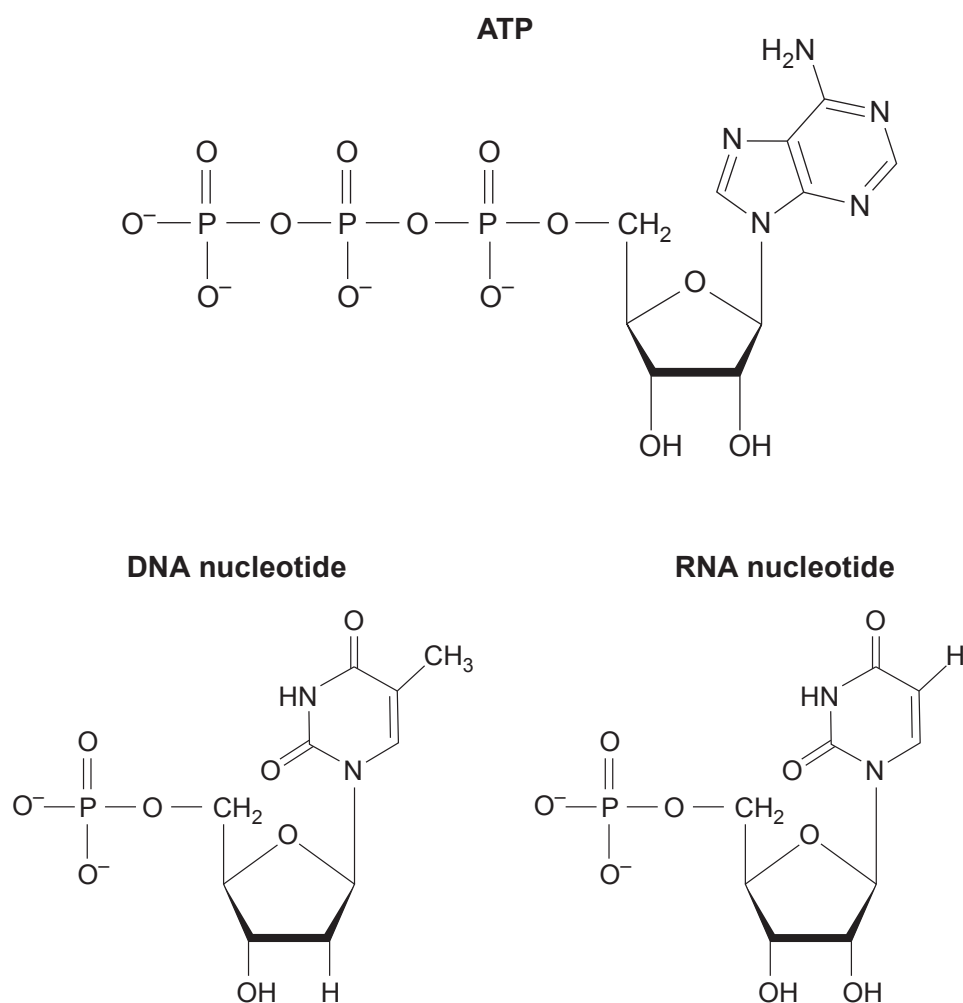


2. Nucleotides are involved in cell metabolism. Three nucleotides are shown in **Image 2.1**.

Image 2.1



- (a) (i) State **three** structural similarities between all three nucleotides shown in **Image 2.1**.

[1]

.....

.....



- (ii) Complete **Table 2.2** to describe **one** structural difference between each of the pairs of nucleotides shown. [2]

Table 2.2

Nucleotides	Difference
ATP and RNA	
DNA and RNA	

- (iii) Name **two** inorganic ions needed by plant cells in order to synthesise the molecules shown in **Image 2.1**. [1]

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- (b) (i) Each human body cell contains 2.6×10^{-11} g of nucleic acid, of which approximately 2.0×10^{-11} g is RNA. Calculate the percentage of nucleic acid that is RNA in a human cell. **Give your answer to one decimal place.** [2]

Percentage of nucleic acid that is RNA =

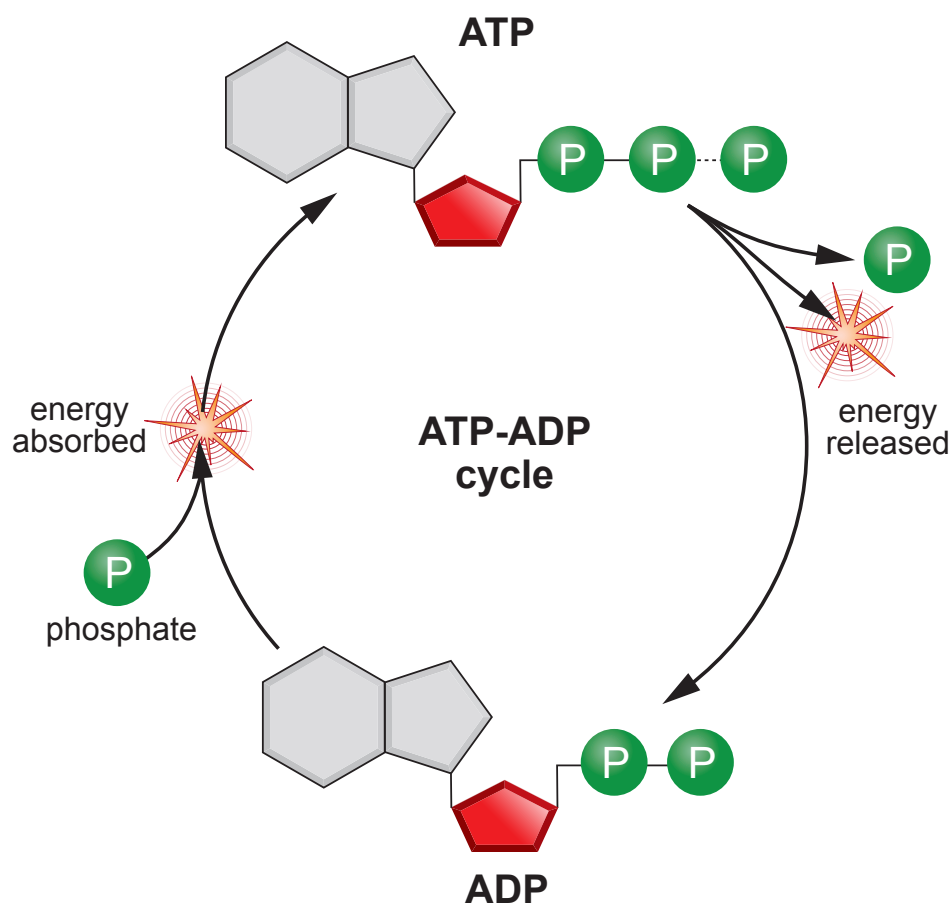
- (ii) Explain why the percentage of nucleic acid that is RNA in a human cell is variable. [1]

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(c) **Image 2.3** shows how ATP is used in a cell.

Image 2.3



Use **Image 2.3** to explain how ATP is used in cells **and** state why it is described as the **universal energy currency**. [3]

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